

Tribute to Rudolf E. Kalman

quest for algebraic
characterizations of positivity
& personal reminiscences

Tryphon Georgiou
Univ. of California, Irvine
IFAC, Toulouse, July 2017

A tribute to

Rudolf E. Kalman



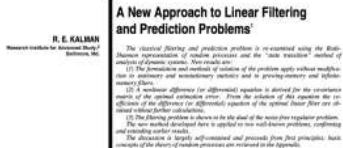
May 19, 1930 - July 2, 2016

The early (miracle) years



A new approach...
20,000+ citations

Kalman, Kalman filtering
~ 1,000,000 citations



Introduction

A significant class of theoretical and practical problems in communication and control is of a statistical nature. Such problems are: (1) Prediction of random signals; (2) separation of random signals from random noise; (3) detection of signals of known form (codes, sinusoids) in the presence of random noise.

In his pioneering work, Wiener [1] showed that problems (1) and (2) can be solved by the so-called Wiener filter. This approach to the problem gave a method (spectral factorization) for the solution of the integral equation in the practically important special case of stationary random signals and stationary spectra.

Many extensions and generalizations followed Wiener's basic work. Zadeh and Ragazzini solved the finite-memory case [2]. Consequently and independently of Brute and Shanon [3], they also gave a simplified method [4] of solution. Brooker discussed the nonstationary Wiener filter equation [5]. These results are now in standard texts [6-8]. A somewhat different approach along these same lines has been given recently by Dirlinger [9]. For extensions to sampled signals, see, e.g., Franklin [10], Loo [11]. Another approach based on the eigenfunctions of the Wiener filter equation certainly applies also to nonstationary problems whereas the preceding methods in general do not. It has been planned by Grier [12] and applied by many others, e.g., Shanon [13], Bode [14], Papoulis [15], Schneiderman [16].

In all these works, the objective is to obtain the specification of a linear dynamic system (Wiener filter) which accomplishes the prediction, separation, or detection of a random signal.²

¹ This research was supported in part by the U. S. Air Force Office of Scientific Research under Contract AF 33(616)-102.

² Bellman Area.

³ Shanon is known to design Bellman's and not of paper.

⁴ It is shown in general that these tasks may be done better by nonlinear filters. At present, however, there is no known method to obtain them theoretically and practically from nonlinear filters.

⁵ Contributions by the Bellman and Ragazzini papers and presented at the Symposium and Symposium Conference, March 28-April 5, 1959, at the Institute for Advanced Study, University of Pennsylvania.

⁶ See, for example, and references therein, papers are to be understood as individual expressions of their authors and not those of the ASME. Manuscript received at ASME Headquarters, February 23, 1960. Paper No. 60-102.

Present methods for solving the Wiener problems are subject to a number of limitations which seriously curtail their practical application:

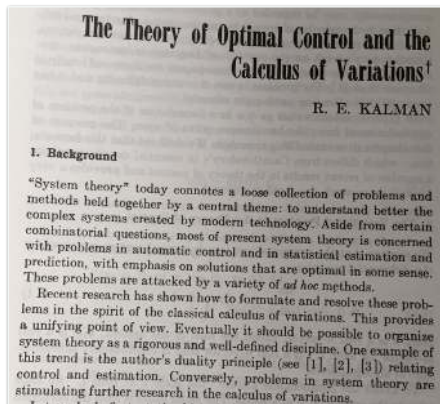
- (1) The optimal filter is specified by its impulse response. It is not a simple task to synthesize the filter from such data.
- (2) Statistical derivation of the optimal impulse response is often quite involved and poorly suited to machine computation. The situation gets rapidly worse with increasing complexity of the problem.
- (3) Important generalizations (e.g., growing-memory filters, nonstationary prediction) require new derivations, frequently of considerable difficulty to the nonmathematician.
- (4) The mathematics of the derivations are not transparent, fundamental assumptions and their consequences tend to be obscured.

This paper introduces a new look at this whole assemblage of problems, clarifying the difficulties just mentioned. The following are the highlights of the paper:

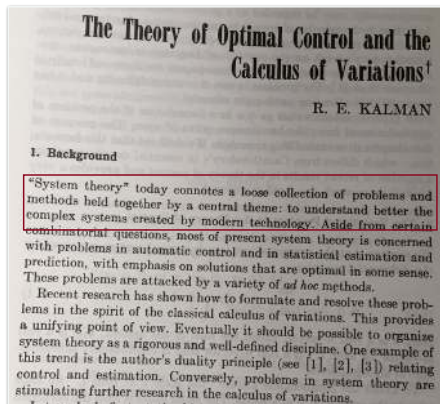
- (1) *Optimal Estimates and Orthogonal Projections.* The Wiener problem is approached from the point of view of conditional distribution and expectations. In this way, basic facts of Wiener theory are quickly obtained; the scope of the results and the fundamental assumptions appear clearly. It is seen that all statistical calculations and results are based on first and second order statistics; no other statistical data are needed. Thus, difficulty (4) is eliminated. This method is well known in probability theory (see pp. 75-78 and 108-115 of Dine [17]) and pp. 402-404 of Loeve [18]) but has not yet been used extensively in engineering.
- (2) *Methods for Random Processes.* Following, in particular, Shanon and Shannon [13], arbitrary random signals are represented (up to second order average statistical properties) as the output of a linear dynamic system excited by independent or uncorrelated random signals ("white noise"). This is a standard trick in the engineering applications of the Wiener theory [17]. The approach taken here differs from the conventional one only in the way in which linear dynamic systems are described. We shall emphasize the concepts of state and state transition; in other words, linear systems will be specified by systems of first-order difference (or differential) equations. This point of view is

Translations of the ASME—Journal of Basic Engineering, 82 (Series D), 38-48, Copyright © 1960 by ASME

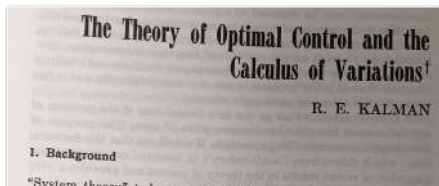
The early (miracle) years



The early (miracle) years



The early (miracle) years



method
comple
combin
with p
predict
These p
Rece
lems in
a unifi
system
this tre
control
stimulat

CONTRIBUTIONS TO THE THEORY OF OPTIMAL CONTROL

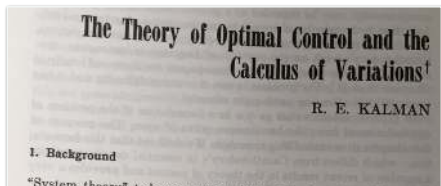
BY R. E. KALMAN

1. Introduction

The purpose of this paper is to give an account of recent research on a classical problem in the theory of control: the design of linear control systems so as to minimize the integral of a quadratic function evaluated along motions of the system. This problem dates back in its modern form to Wiener and Hall at about 1943 ([1], [2]). In spite of its relatively long history, the problem has never been formulated rigorously from a mathematical point of view. Even the most up-to-date expositions of the subject (see, e.g., [3]) are inaccessible to the mathematician due to the lack of precisely stated conditions and results.

The problem is quite broad, and there are many unsettled questions. This

The early (miracle) years



method
comple
combin
with p
predict
These p
Rece
lems in
a unifi
system
this tre
control
stimulat

CONTRIBUTIONS TO THE THEORY OF OPTIMAL CONTROL

BY R. E. KALMAN

1. Introduction

The purpose of this paper is to give an account of recent research on a classical problem in the theory of control: the design of linear control systems so as to minimize the integral of a quadratic function evaluated along motions of the system. This problem dates back in its modern form to Wiener and Hall at about 1943 ([1], [2]). In spite of its relatively long history, the problem has never been formulated rigorously from a mathematical point of view. Even the most up-to-date expositions of the subject (see, e.g., [3]) are inaccessible to the mathematician due to the lack of precisely stated conditions and results.

The problem is quite broad, and there are many unsettled questions. This

The early (miracle) years

The Theory of Optimal Control and the Calculus of Variations[†]

R. E. KALMAN

1. Background

"System theory" today connotes a loose collection of problems and methods held together by a central theme: to understand better the complex systems created by modern technology. Aside from certain combinatorial questions, most of present system theory is concerned with problems of optimization.

predict

These

Rece

lems in

a unify

system

this tre

control

stimula

CONTRIBUTIONS TO THE THEORY OF OPTIMAL CONTROL

BY R. E. KALMAN

1. Introduction

The purpose of this book is to present the problem in the theory of optimal control. This problem was first formulated in 1943 ([1], [2]). The problem was then reformulated in a more rigorous manner by Pontryagin and his co-workers in 1947. The problem was then reformulated in a more rigorous manner by Pontryagin and his co-workers in 1947.

The problem

R. E. KALMAN

Algebraic Theory of Linear Systems*

1. INTRODUCTION

Elementary textbooks in system theory present the Laplace transform as a magic and mysterious gadget. Somehow the difficult problem (?) of solving linear differential equations with constant coefficients is replaced with the easy problem (?) of factoring polynomials and computing partial fractions.

The early (miracle) years

The Theory of Optimal Control and the Calculus of Variations[†]

R. E. KALMAN

1. Background

"System theory" today connotes a loose collection of problems and methods held together by a central theme: to understand better the complex systems created by modern technology. Aside from certain combinatorial questions, most of present system theory is concerned with problems of optimization.

predict

These

Rece

lems in

a unify

system

this tre

control

stimula

CONTRIBUTIONS TO THE THEORY OF OPTIMAL CONTROL

BY R. E. KALMAN

1. Introduction

The purpose of this book is to present the problem in the theory of optimal control. This problem was first formulated in 1943 ([1], [2]). The problem was formulated in a rigorous manner by Pontryagin and his colleagues in 1947. The problem was then formulated in a rigorous manner by Pontryagin and his colleagues in 1947.

The problem

R. E. KALMAN

Algebraic Theory of Linear Systems*

1. INTRODUCTION

Elementary textbooks in system theory present the Laplace transform as a magic and mysterious gadget. Somehow the difficult problem (?) of solving linear differential equations with constant coefficients is replaced with the easy problem (?) of factoring polynomials and computing partial fractions.

The early (miracle) years

The Theory of Optimal Control and the Calculus of Variations†

I. Background

"System theory" methods held to complex systems combinatorial qu with problem predict These Rec lems in a unify system this tre control stimulat

The problem mini

system. This p 1943 ([1], [2]). formulated rig date exposition tician due to t The problem

R. E. KALMAN

Research Institute for Advanced Studies
(RIAS), Baltimore, Md.

When Is a Linear Control System Optimal?

The purpose of this paper is to formalize, study, and (in certain cases) resolve the Inverse Problem of Optimal Control Theory, which is the following: Given a control law, find all performance indices for which this control law is optimal.

Under the assumption of (a) linear constant plant, (b) linear constant control law, (c) measurable state variables, (d) quadratic loss functions with constant coefficients, (e) single control variable, we give a complete analysis of this problem and obtain various explicit conditions for the optimality of a given control law. An interesting feature of the analysis is the central role of frequency-domain concepts, which have been ignored in optimal control theory until very recently. The discussion is presented in rigorous mathematical form.

The central conclusion is the following (Theorem 6): A stable control law is optimal if and only if the absolute value of the corresponding return difference is at least equal to one at all frequencies. This provides a beautifully simple connecting link between modern control theory and the classical point of view which regards feedback as a means of reducing component variations.

Introduction

THE UNDER interest in optimal control theory in recent years has brought with it a good deal of confusion.

There are nagging doubts—particularly strongly felt by engineers—that the theory is overidealized, hence impractical. It is argued that the theory of the performance index to be optimized is arbitrary and subjective, perhaps only a matter of taste. If so, then it is pointless to devote too much effort to finding a control law which is the best in some narrow, individualistic sense—it would be more sensible to seek approximate control laws which are not rigidly tied to a single performance index.

- (a) All state variables can be directly measured.
- (b) The loss functions are quadratic forms with constant coefficients in the state and control variables.
- (c) There is only one control variable.

The first four of these assumptions are made in order to obtain explicit results. The last assumption can be removed, but only at the cost of a more tedious analysis [2].

It is hoped that this paper will reduce the "gap" which exists today between optimal control theory and conventional control engineering practice.

There are many interesting and unexpected results. Let us

Algebraic Theory of Linear Systems*

1. INTRODUCTION

Elementary textbooks in system theory present the Laplace transform as a magic and mysterious gadget. Somehow the difficult problem (?) of solving linear differential equations with constant coefficients is replaced with the easy problem (?) of factoring polynomials and computing partial fractions.

The early (miracle) years

LYAPUNOV FUNCTIONS FOR THE PROBLEM OF LUR'E IN AUTOMATIC CONTROL*

By R. E. KALMAN

RESEARCH INSTITUTE FOR ADVANCED STUDY (RIAS), BALTIMORE, MD.

Communicated by S. Lefschetz, December 18, 1962

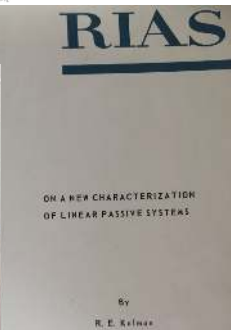
1. About 1950, Lur'e¹ initiated the study of a class of (closed-loop) control systems whose governing equations are

$$dx/dt = Fx - p\varphi(\sigma), \quad d\sigma/dt = -\varphi(\sigma), \quad \sigma = K'x + p\dot{x}. \quad (1)$$

In (1), σ , t , p are real scalars, K , p , λ are real. The prime denotes the transpose. F is a real matrix. $\varphi(\sigma)$ is a real-valued, continuous A_1 ; $\varphi(0) = 0$, $0 < \sigma\varphi(\sigma) < \sigma^2$.

We ask: Is the equilibrium point $x = 0$, $\sigma = 0$ stable for any $\varphi \in A_1$?

2. This problem is $\{1\}$ is *g.a.v.* for every λ *crude form*, however, had to consider a more



The early (miracle) years

Least Squares Stationary Optimal Control and the Algebraic Riccati Equation

ISSN 0013-788X, 1969, vol. 1, no. 1, pp. 1-10.

Abstract. The classical control of linear systems with respect to quadratic performance is discussed in the frequency domain. The algebraic Riccati equation is derived from the least squares stationary optimal control problem. The algebraic Riccati equation is shown to be equivalent to the frequency domain least squares stationary optimal control problem.

Although the least squares optimization problem with linear differential constraints has roots going back to the very beginnings of calculus of variations, its revival and introduction in control theory may safely be credited to Kalman [1]. We should also mention Newton et al. [2], who put forward least squares techniques as a systematic basis for the design of stationary feedback control systems. Many of the results of this paper are inspired by some of the results obtained by Brockett. His work has appeared in various places in the literature and may be found in summarized form in the recent text [3]. Other papers which make ample contact with the results presented here are some of the later papers by Kalman [4], [5], the important paper by Popov [6], and the work of Anderson (see, for example, Anderson [7] and Anderson and Moore [8]).

The analytical treatment of the class of optimization problems introduced in the preceding leads to a series of matrix relations and frequency domain inequalities. Those that are important for our purposes are listed as follows. We will be interested in the case $K = K^*$.

1) The Linear Matrix Inequality (LMI):

$$F(K) = \begin{bmatrix} A^*K + KA + Q & KB + C^* \\ B^*K + C & R \end{bmatrix} \geq 0.$$

2) The Quadratic Matrix Inequality (QMI):

$$A^*K + KA - (KB + C^*)R^{-1}(B^*K + C) + Q \geq 0.$$

3) The Algebraic Riccati Equation (ARE):

$$A^*K + KA - (KB + C^*)R^{-1}(B^*K + C) + Q = 0.$$

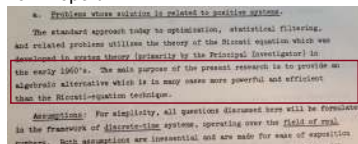
4) The Frequency-Domain Inequality (FDI):

$$H(i\omega) = R + C(I\omega - A)^{-1}B + B^*(I\omega - A^*)^{-1}C^* \\ + B^*(I\omega - A^*)^{-1}Q(I\omega - A)^{-1}B \geq 0.$$

It is very well known that the ARE plays a crucial role in the solution of the optimal control problem under consideration. (One often gets the impression that this equation in fact constitutes the bottleneck of all of linear system theory.) However, it is much less appreciated how the other relations enter into the theory. We hope that their role will be clarified in this paper.

The seventies: realization with positivity

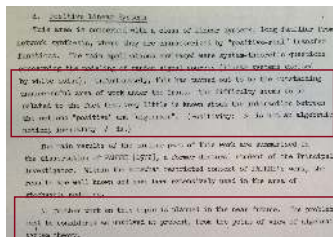
1972 report



Since a linear discrete-time system is completely determined by a pair of matrices whose behavior is identical with that of its discrete-time.

$$\left\{ \begin{array}{l} \text{system} \\ \text{state} \\ \text{output} \end{array} \right\} = \left\{ \begin{array}{l} \text{matrix} \\ \text{matrix} \\ \text{matrix} \end{array} \right\} \rightarrow \left\{ \begin{array}{l} \text{behavior} \\ \text{matrix} \\ \text{matrix} \end{array} \right\} \rightarrow \left\{ \begin{array}{l} \text{matrix} \\ \text{matrix} \\ \text{matrix} \end{array} \right\}$$

1977 report



Problem:

- characterize minimal system realizations that are also passive

Partial realization with positivity

Realization of covariance sequences:

suppose $\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_n$ are the autocorrelation lags of a stationary (zero-mean) discrete time stochastic process, i.e.,

$$\mathbf{c}_\ell = \mathbb{E}\{\mathbf{y}(k)\mathbf{y}(k + \ell)\}.$$

Characterize possible extensions $\mathbf{c}_{n+1}, \mathbf{c}_{n+2}, \dots$ such that

i) the series

$$F(z) := \frac{1}{2} \mathbf{c}_0 + \mathbf{c}_1 z^1 + \mathbf{c}_2 z^2 + \dots + \mathbf{c}_n z^n + \mathbf{c}_{n+1} z^{n+1} + \dots$$

converges in $|z| < 1$ and positive real part (positive-real function)

ii) $F(z)$ defines a rational function of minimal degree.

Partial realization with positivity

Realization of circuits from transfer function data:

suppose $w_0, w_1, \dots, w_n$ denote the values that the transfer function $F(z)$ takes at points $z_0, z_1, \dots, z_n$ in $\{z \mid |z| < 1\}$.

Characterize all rational functions $F(z)$, analytic with positive real part in $|z| < 1$, such that

i)

$$F(z_i) := w_i \text{ for all } i \in \{0, 1, \dots, n\}.$$

ii) $F(z)$ is rational function and of minimal degree.

Partial realization with positivity

Necessary & sufficient condition (existence of solutions):

$$\operatorname{Re} F(S) \geq 0$$

where S is the “compressed shift”, i.e., multiplication by z^1 restricted on $H_2 \ominus B(z)H_2$

$$B(z) = z^n, \text{ or } \prod_{i=0}^n \frac{z - z_i}{1 - \bar{z}_i z}$$

translates to:

a Toeplitz (Pick) matrix being positive semi-definite (respectively)

Will be assumed throughout

Partial realization with positivity

- **Analytic interpolation with a degree constraint**

Kalman argues that the complexity of the answer from an engineering standpoint may not be the same as that sought by physicists, information theorists, etc.

- **Applications:**

stochastic identification (Faurre, ...)

inverse problems (recently, Baratchart, ...)

circuit synthesis

Partial realization with positivity

REALIZATION OF COVARIANCE SEQUENCES

2004年12月10日

This paper examines the problem of "positivity" in relation to the partial realization of scalar power series. An exact criterion of positivity is proved for second-order realizations. The general case is currently unresolved. Even the special results contained here show that the so-called "maximum entropy principle" cannot be applied to the realization problem in the naive sense in which it is employed by physicists. It would be better to call this principle a "precaution" because it does not fully utilize the information inherent in the data and hence not properly a realization with natural ("minimal") mathematical properties.

The research I planned to report on is unfortunately not yet completed. So the following is only an outline of the problem. It has been around for a long time without receiving a definitive solution. It occupies quite a central position in system theory and has been frequently misinterpreted. It is not in the least controversial, but it is unsolved.

As perhaps the only slightest at this meeting, it is safer for me if I use a reasonably nontechnical language. In such terms, the topic of my paper is: What is the relation between partial realizations and primitivity?

Evidently I must now define "partial realization" and (in relation to it) "positivity".

To take the simplest (scalar) case, consider an arbitrary sequence of numbers from a fixed field K .

* Presented at Topolite Workshop Conference at Tel Aviv University, Tel Aviv, ISRAEL, on May 12, 1981. This research was supported in part under US Air Force Grant AFOSR 76-3034(D) and US Army Research Grant DAK 25-77-G-0225 through the Center for Mathematical System Theory, University of Florida.

But (25) is not sufficient to insure that the infinite sequence generated by (17) is positive! The necessary and sufficient condition for the latter requirement is (after rather extensive calculations) found to be

$$(21) \quad |x_3| = \frac{|x_2|}{1 + |x_2|} \leq \frac{1}{2}$$

that the $\{x_n\}$ is a partial realization of the $\{x_n\}$ is a positive partial realization of the $\{x_n\}$ if $x_n \geq 0$ for all n .

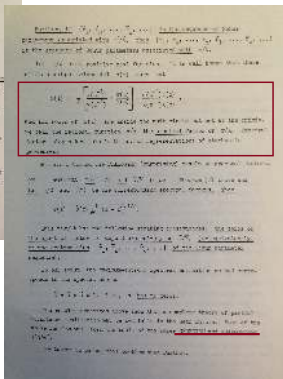
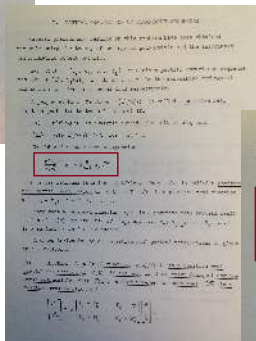
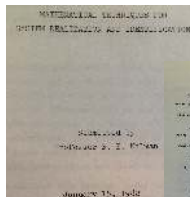
[illegible][illegible]

7. The above is repeated with the following additional points:
Each time before the meeting given to the staff, the Chairman of
JASTING and his staff is expected to be in the room and should
submit a statement. It should be given to the staff in the
presence of the staff and the Chairman of JASTING.

how does minimality square with uniqueness?

Since β is an unbounded dyadic field, β^* has a unique
maximal ideal which is identical with that of its closure.

The eighties - pressing on



Caratheodory & Nevanlinna-Pick Interpolation

– Early insights/classical

Parametrization of all solutions via an LFT

(Linear Fractional Transformation) Q : positive real

$$F(z) = \text{LFT}_{\text{data}}(Q) \quad \text{i.e.,} = \frac{A + BQ}{C + DQ}$$

Caratheodory & Nevanlinna-Pick Interpolation

- **Early insights/classical**

Parametrization of all solutions via an LFT
(Linear Fractional Transformation) Q : positive real

$$F(z) = \text{LFT}_{\text{data}}(Q) \quad \text{i.e.,} = \frac{A + BQ}{C + DQ}$$

$$\text{Re}F(e^{j\theta}) \sim \text{Re}Q(e^{j\theta})$$

Caratheodory & Nevanlinna-Pick Interpolation

– Classical/early insights

Parametrization of all solutions via an LFT

(Linear Fractional Transformation) Q : positive real

$$F(z) = \text{LFT}_{\text{data}}(Q) \quad \text{i.e.,} = \frac{A + BQ}{C + DQ}$$

$$\text{Re}F(e^{j\theta}) \sim \text{Re}Q(e^{j\theta})$$

also,

$$F \text{ rational} \Leftrightarrow Q \text{ rational}$$

Georgiou-Khargonekar 1981

Caratheodory & Nevanlinna-Pick Interpolation

– Classical/early insights

Parametrization of all solutions via an LFT

(Linear Fractional Transformation) Q : positive real

$$F(z) = \text{LFT}_{\text{data}}(Q) \quad \text{i.e.,} = \frac{A + BQ}{C + DQ}$$

However,

$$|n - \deg Q| \leq \deg F \leq n + \deg Q$$

How to choose Q ?

Algebraic characterization unknown?

Analytic Interpolation with degree constraint

- **Topological constraints (Georgiou 1983, 1987)**

Degree n solutions via Homotopy methods

For any polynomial η of degree $\leq n$,
there exists F of degree n , i.e., $F = \pi/\chi$,
 π, χ polynomials of degree $\leq n$, such that

$$\operatorname{Re} F(z) = \frac{|\eta(z)|^2}{|\chi(z)|^2} \text{ for } z = e^{i\theta}$$

is positive real and satisfies $n + 1$ interpolation constraints

η : “spectral numerator”

moving average part of an f.d. stochastic realization

Analytic Interpolation with degree constraint

- **Generic cases (Georgiou 1983):**

Both data sets:

$$\{c'_i s \text{ or } w'_i s \mid \text{minimal realization has degree} = n\}$$

as well as the opposite,

$$\{c'_i s \text{ or } w'_i s \mid \text{minimal realization has degree} < n\}$$

have open interiors (generic).

$\Rightarrow$ **Kalman's problem has a non-trivial set of solutions**

1990's & 2000's - state of the problem

Via optimization & other methods:

Complete parametrization

$$\{\text{solutions of degree } \leq n\} \leftrightarrow \{\eta \mid \text{stable of degree } \leq n\}$$

Byrnes-Lindquist-Gusev-Matveev 1995 (uniqueness)

Byrnes-Landau-Lindquist 1997

Burnes-Gusev-Lindquist 1998 (KL-optimization)

Georgiou 1999

Byrnes-Georgiou-Lindquist 2001, 2003,...

Byrnes-Georgiou-Lindquist-Megretski 2006

Takyar-Georgiou 2007 (matrix-valued)

1990's & 2000's - state of the problem

Byrnes-Georgiou-Lindquist-Megretski (Trans. AMS, 2004)

Generalized interpolation in H^∞ with a complexity constraint

$K := H_2 \ominus B(z)H_2$ with B arbitrary inner (i.e., B is “all-pass”)
 T bounded operator on K , commutes with S

$\operatorname{Re}(T) \geq 0$ (necessary condition)

$\exists$ solutions $F = \pi/\chi$ with $\pi, \chi \in K$, i.e., $F(S) = T$

any such solution $\leftrightarrow \eta \in K$ via $\operatorname{Re} F = \frac{|\eta|^2}{|\chi|^2}$

Further F is the unique maximizer of

$$F \mapsto \int |\eta|^2 \log(\operatorname{Re} F)$$

state of the problem

- **Complete parametrization of degree $\leq n$ solutions**

Smooth parametrization $\leftrightarrow$ stable spectral numerator

Multivariable interpolation (**F** matrix-valued)

Applications (spectral analysis, inverse problems)

Kalman's problem ($< n?$) remains open:
algebraic characterization of minimal degree
positive-real interpolants

when is the minimal degree $< n$?

what is the minimal degree?

are there suitable algebraic invariants?

A tribute to Professor Rudolf E. Kalman



Gainesville, June 12, 2016